

Transportation Optimization

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Modified Kruskal Algorithm for Infrastructure Network Design

Prepared by: Ibrahim Mohamed Anwar

Introduction:

Kruskal is a greedy algorithm used to find a Minimum Spanning Tree (MST).

MST connects all nodes with minimum total cost.

- Used in:
- Road planning
- Electrical networks
- Communication systems
- In transportation:
- Helps design efficient road infrastructure.

Modified Kruskal Algorithm for Infrastructure Network Design

Prepared by: Ibrahim Mohamed Anwar

Mathematical Foundation

Graph representation: $G=(V,E)$

- V = intersections/facilities
- E = roads

MST OBJECTIVE:

$\min \sum_{e \in T} w(e)$

- Kruskal selects smallest edges without creating cycles.

PSEUDOCODE:

- Sort all edges by weight.
- Start with empty tree.
- Add smallest edge if no cycle forms.
- Repeat until all vertices are connected.

COMPLEXITY ANALYSIS:

Sorting edges:

$O(E \log E)$

Union-Find operations:

$O(E \alpha(V))$

Total complexity:

$O(E \log E)$

Modified Kruskal Algorithm for Infrastructure Network Design

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Use in This Transportation Scenario

- Roads are modeled as graph edges.
- Important locations are graph nodes.
- Existing roads receive lower costs.
- Congested or damaged roads receive higher weights.
- Airport and hospitals are prioritized.
- The algorithm helps determine which roads should be connected first with minimum construction cost.

Performance Analysis

- Efficient for large sparse road networks.
- Quickly builds low-cost infrastructure layouts.
- Reduced total construction cost.
- Maintains connectivity between all districts.

Comparison with Alternatives

Algorithm	Advantage	Disadvantage
Kruskal	Simple and efficient	Less suitable for dense graphs
Prim	Better for dense graphs	Needs adjacency structures
Steiner Tree	More realistic	Much more complex

Modified Kruskal Algorithm for Infrastructure Network Design

Prepared by: Ibrahim Mohamed Anwar

Conclusion:

- Kruskal provides efficient infrastructure planning.
- Greedy strategy gives near-optimal road layouts.
- Transportation modifications improve realism.
- Good balance between simplicity and performance.

Dijkstra's Algorithm for Standard Route Planning

Prepared by: Moaz Mahmoud Gaber

Introduction:

Dijkstra computes shortest paths from one source.

Works with nonnegative edge weights.

Used in:

- GPS systems
- Navigation apps
- Transportation routing

Dijkstra's Algorithm for Standard Route Planning

Prepared by: Moaz Mahmoud Gaber

Mathematical Foundation

$\text{dist}(v) = \min(\text{dist}(v), \text{dist}(u) + w(u, v))$

- Relaxation updates shortest distances.
- Priority queue selects minimum distance node.

PSEUDOCODE:

- Initialize distances.
- Insert source into priority queue.
- Extract smallest distance node.
- Relax neighboring edges.
- Repeat until queue is empty.

COMPLEXITY ANALYSIS:

$O((V+E)\log V)$

Dijkstra’s Algorithm for Standard Route Planning

Prepared by: Moaz Mahmoud Gaber

Use in This Transportation Scenario

- Nodes represent locations in Cairo.
- Edge weights represent travel time.
- Congestion increases edge cost.
- Closed roads are ignored.
- Finds fastest routes between districts.

Performance Analysis

- Produces accurate shortest routes.
- Fast for city-scale routing.
- Handles changing traffic weights efficiently.

Comparison with Alternatives

Algorithm	Advantage	Disadvantage
Dijkstra	Accurate shortest path	No heuristic guidance
Bellman–Ford	Handles negative weights	Slower
A*	Faster with heuristic	Needs good heuristic

Dijkstra's Algorithm for Standard Route Planning

Prepared by: Moaz Mahmoud Gaber

Conclusion:

- Dijkstra is reliable for urban navigation.
- Efficient for transportation systems.
- Easy to adapt for congestion-aware routing.

A* Search for Emergency Vehicle Routing

Prepared by: Omar Tamer Abdullah

Introduction:

A* is a heuristic shortest path algorithm.

Combines actual and estimated costs.

Common in:

- Emergency response
- Robotics
- GPS navigation

A* Search for Emergency Vehicle Routing

Prepared by: Omar Tamer Abdullah

Mathematical Foundation

$$f(n) = g(n) + h(n)$$

- $g(n)$ = actual path cost
- $h(n)$ = estimated remaining cost

PSEUDOCODE:

- Add source to open list.
- Select node with smallest $f(n)$.
- Expand neighbors.
- Update costs.
- Stop when destination reached.

COMPLEXITY ANALYSIS:

$$O((V+E)\log V)O((V+E)\log V)O((V+E)\log V)$$

A* Search for Emergency Vehicle Routing

Prepared by: Omar Tamer Abdullah

Use in This Transportation Scenario

- Used for ambulances and emergency vehicles.
- Euclidean distance used as heuristic.
- Congestion penalties reduced for emergency vehicles.
- Finds fastest emergency route to hospitals.

Performance Analysis

- Explores fewer nodes than Dijkstra.
- Faster response times.
- Efficient in urgent scenarios.

Comparison with Alternatives

Algorithm	Advantage	Disadvantage
A*	Faster search	Needs heuristic
Dijkstra	Guaranteed shortest path	Slower
Greedy BFS	Very fast	Not always optimal

A* Search for Emergency Vehicle Routing

Prepared by: Omar Tamer Abdullah

Conclusion:

- A* improves emergency response efficiency.
- Heuristic guidance reduces search time.
- Very effective for real-time routing.

Time-Dependent Shortest Path Algorithm

Prepared by: Ahmed Ibrahim Mohamed

Introduction:

- Traditional routing assumes constant edge weights.
- Real traffic changes over time.
- Time-dependent routing models congestion dynamically.

Time-Dependent Shortest Path Algorithm

Prepared by: Ahmed Ibrahim Mohamed

Mathematical Foundation

$w(e,t)$

- Edge cost depends on time.

FIFO property:

- Leaving earlier should not arrive later.

PSEUDOCODE:

- Determine current time slot.
- Update edge weights.
- Apply shortest path search.
- Recalculate during traffic changes.

COMPLEXITY ANALYSIS:

$O((V+E)\log V)$

Time-Dependent Shortest Path Algorithm

Prepared by: Ahmed Ibrahim Mohamed

Use in This Transportation Scenario

- Different weights for:
- Morning
- Afternoon
- Evening
- Night
- Congestion increases travel time.
- More realistic for Cairo traffic conditions.

Performance Analysis

- Produces more realistic routes.
- Adapts to rush-hour traffic.
- Improves travel-time estimation accuracy.

Comparison with Alternatives

Algorithm	Advantage	Disadvantage
Time-Dependent	Realistic traffic modeling	More complex
Static Dijkstra	Simpler	Ignores congestion
Simulation Models	Highly accurate	Computationally expensive

Time-Dependent Shortest Path Algorithm

Prepared by: Ahmed Ibrahim Mohamed

Conclusion:

- Time-aware routing improves transportation realism.
- Better for congested urban environments.

Dynamic Programming for Public Transportation Scheduling

Prepared by: Mohamed Ahmed Abdeltawab

Mathematical Foundation

$$DP(i,b)=\max_{x\in[0,b]}(\text{served}(i,x)+DP(i+1,b-x))$$

Dynamic Programming for Public Transportation Scheduling

Prepared by: Mohamed Ahmed Abdeltawab

Use in This Transportation Scenario

- Determines how extra buses are distributed.
- Maximizes transported passengers.
- Routes with higher demand receive priority.
- Helps reduce overcrowding.

Comparison with Alternatives

Algorithm	Advantage	Disadvantage
Dynamic Programming	Optimal allocation	Higher memory usage
Greedy Allocation	Fast	Not always optimal
Brute Force	Exact	Very slow

Greedy Traffic Signal Optimization

Prepared by: Mohamed Islam

Introduction:

Greedy algorithms make best immediate decisions.

Used in:

- Traffic lights
- Scheduling
- Resource allocation

Greedy Traffic Signal Optimization

Prepared by: Mohamed Islam

Use in This Transportation Scenario

- Green time assigned according to queue size.
- Congested roads receive longer green signals.
- Minimum green time prevents starvation.
- Supports real-time traffic management.

Comparison with Alternatives

Algorithm	Advantage	Disadvantage
Greedy	Fast and simple	Not globally optimal
Dynamic Optimization	More accurate	Slower
Reinforcement Learning	Adaptive	Complex

Emergency Vehicle Preemption System

Prepared by: Yassin Nader Ahmed Ibrahim

Introduction:

Emergency preemption gives priority to emergency vehicles.

Used for:

- Ambulances
- Fire trucks
- Police vehicles

Emergency Vehicle Preemption System

Prepared by: Yassin Nader Ahmed Ibrahim

Mathematical Foundation

Priority score based on congestion severity and emergency importance.

PSEUDOCODE:

- Detect emergency vehicle.
- Identify nearest intersections.
- Prioritize signal changes.
- Clear path dynamically.

COMPLEXITY ANALYSIS:

$O(k \log k)$

Emergency Vehicle Preemption System

Prepared by: Yassin Nader Ahmed Ibrahim

Use in This Transportation Scenario

- Traffic signals automatically adjust for ambulances.
- Congested intersections are cleared first.
- Reduces emergency response delay.
- Improves survival chances in critical cases.

Performance Analysis

- Faster emergency travel time.
- Reduced intersection waiting.
- Improved traffic responsiveness.

Comparison with Alternatives

Algorithm	Advantage	Disadvantage
Emergency Preemption	Immediate response	Can disturb normal traffic
Fixed-Time Signals	Simple	Not adaptive
AI-based Systems	More intelligent	More expensive

Emergency Vehicle Preemption System

Prepared by: Yassin Nader Ahmed Ibrahim

Conclusion:

- Emergency preemption improves public safety.
- Real-time control is essential in smart transportation systems.
- Effective for highly congested cities like Cairo.

Thankyou

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